Performance Report – ATS Optimization

1. Execution Speed Improvement

- Implemented resume caching to avoid repeated file reading
- Reduced disk I/O operations

2. Similarity Optimization

- Introduced similarity caching to prevent recomputation
- Reduced model computation time significantly

3. Data Processing Optimization

- Applied text normalization and cleaning
- Improved input quality for matching

4. Memory Optimization

- Processed resumes once and reused across JDs
- Reduced redundant memory usage

5. Scalability

- System supports multiple job descriptions (65 JDs)
- Handles large number of resumes efficiently (100+)

Conclusion: The optimized ATS engine significantly improves execution speed, reduces redundancy, and supports scalable candidate evaluation.